Seizure

UP-13 | Universal Patient Care

Quick flow

01 Protect + Assess [EMT / AEMT / Paramedic]

Prevent injury • check pulse • airway position • suction • oxygen or BVM when indicated

02 Status Epilepticus? [EMT / AEMT / Paramedic]

Generalized seizure ≥5 minutes • unknown prolonged duration • recurrent without recovery

03 Glucose + Reversible Causes [EMT / AEMT / Paramedic]

Check glucose without delaying anticonvulsant • treat hypoglycemia under UP-4

04 Midazolam [Paramedic]

Adult: 10 mg IM/IN or 5 mg IV/IO • Pediatric: weight-based • maximum total 10 mg

05 Still Seizing? [Paramedic / Medical Control]

Adult: ketamine 1 mg/kg IV/IO/IM, max 100 mg • Pediatric: contact Medical Control

06 Continuous Monitoring [EMT / AEMT / Paramedic]

Cardiac monitor • SpO_2 • waveform $EtCO_2$ after medication • airway and BVM readiness

07 Reassess + Transport [EMT / AEMT / Paramedic]

Neurologic recovery • trauma • pregnancy/postpartum • fever • toxin • stroke • document times and doses

Overview

- Prioritize patient protection, airway and ventilation support, rapid recognition of convulsive status epilepticus, timely anticonvulsant administration, and identification of reversible causes.
- Convulsive status epilepticus is continuous generalized seizure activity lasting 5 minutes or longer, or recurrent seizures without recovery to baseline between events.
- Treat the seizure while evaluating for hypoxia, hypoglycemia, cardiac arrest, trauma, stroke, pregnancy or postpartum eclampsia, infection, toxins, withdrawal, and medication nonadherence.

Provider Actions

Treat status epilepticus promptly while supporting oxygenation and ventilation.

EMT

- Protect the patient from injury, loosen restrictive clothing, position laterally when feasible, suction as needed, and provide oxygen or BVM ventilation when clinically indicated.
- Check pulse and rhythm promptly because brief seizure-like activity may occur at the onset of cardiac arrest.
- Obtain glucose without delaying indicated anticonvulsant administration; treat hypoglycemia under UP-4.
- Determine seizure onset, duration, number of events, recovery between events, and medications given before EMS arrival.

AEMT

- Perform all EMT care. Establish IV/IO access when clinically useful, but do not delay IM or IN midazolam while attempting vascular access.
- Treat hypoglycemia under UP-4, assist ventilation, trend vital signs, and support continuous monitoring and rapid transport.

Paramedic

- Perform all prior care. Administer midazolam for convulsive status epilepticus or recurrent seizures without recovery.
- For persistent adult seizure after the adequate midazolam regimen, administer ketamine 1 mg/kg IV/IO or IM, maximum 100 mg.
- For pediatric ketamine, contact Medical Control for authorization and dosing.
- Prepare for respiratory depression, assisted ventilation, and advanced airway management after anticonvulsant medication.

Adult Anticonvulsant Pathway

Use an adequate early dose and do not delay treatment for vascular access.

EMT

- Continue patient protection, airway positioning, suction, oxygenation, ventilation, glucose assessment, and preparation for medication administration.

AEMT

- Obtain IV/IO access when it can be accomplished without delaying anticonvulsant treatment.

Paramedic

- Without established IV/IO access: midazolam 10 mg IM is preferred. Midazolam 10 mg IN, divided between nostrils, is an alternative when IM administration is impractical.
- With established IV/IO access: midazolam 5 mg IV/IO; repeat 5 mg once after 5 minutes if seizure continues. Maximum total EMS midazolam dose is 10 mg.
- If seizure persists after the adequate midazolam regimen: ketamine 1 mg/kg IV/IO or IM once, maximum 100 mg. Continue cardiac monitoring, SpO_2, waveform $EtCO_2$, and immediate airway and ventilation readiness.

Pediatric Anticonvulsant Pathway

Pediatric patients are younger than 16 years; use actual or length-based weight.

EMT

- Use pediatric airway equipment and length-based resources. Identify prescribed rescue medication and the exact dose, route, and time given before EMS arrival.

AEMT

- Perform all EMT care and obtain IV/IO access when it will not delay anticonvulsant treatment.

Paramedic

- Midazolam IM or IN: 0.2 mg/kg, maximum single dose 10 mg.
- Midazolam IV/IO: 0.1 mg/kg, maximum single dose 5 mg.
- May repeat once after 5 minutes if seizure continues. Maximum total midazolam is 0.4 mg/kg, not to exceed 10 mg.
- Count prescribed rescue benzodiazepine administered before EMS arrival toward the maximum of two total benzodiazepine doses during the episode.
- For seizure persisting after the adequate benzodiazepine regimen, contact Medical Control before pediatric ketamine administration.

Indications

- Active generalized or focal seizure.
- Convulsive seizure lasting 5 minutes or longer, unknown prolonged duration, or recurrent seizures without return to baseline.

- Postictal altered mental status requiring evaluation for injury, reversible cause, or incomplete neurologic recovery.

Assessment | History

- Document exact seizure onset time, duration, type, number of seizures, recovery between events, and baseline neurologic status.
- Ask about previous seizures, prescribed antiseizure medications, adherence, recent dose changes, prescribed rescue medication, and all medications administered before EMS arrival.
- Assess for pregnancy and postpartum status, trauma, diabetes, fever or infection, alcohol or sedative withdrawal, medication or toxin exposure, overdose, headache, stroke symptoms, and recent illness.

Assessment | Examination

- Assess airway, ventilation, oxygenation, circulation, pulse, mental status, pupils, gaze or eye deviation, focal movement, symmetry, and postictal neurologic recovery.
- Examine for head or neck injury, tongue or oral trauma, active bleeding, incontinence, hyperthermia, meningismus, rash, and evidence of ingestion or injection.
- Obtain glucose in every patient when feasible. Obtain temperature when fever, infection, hyperthermia, or pediatric febrile seizure is possible.
- Persistent unilateral weakness, aphasia, gaze deviation, or another focal deficit beyond the expected postictal period requires stroke and intracranial-emergency evaluation.

Assessment | Monitoring

- Use continuous cardiac monitoring and SpO_2 for status epilepticus, recurrent seizure, abnormal vital signs, or medication administration.
- Apply waveform $EtCO_2$ after midazolam or ketamine and whenever ventilation is impaired or assisted.
- Obtain a 12-lead ECG when a cardiac cause, syncope, toxin, overdose, electrolyte disturbance, or abnormal rhythm is suspected; it is not required for every uncomplicated seizure.
- Repeat complete vital signs, respiratory assessment, mental status, and neurologic examination after seizure cessation and every medication dose.

Treatment steps

- Protect the patient from injury without forcibly restraining convulsions. Never place an object, medication, finger, or bite block in the mouth during active seizure.
- Position the airway, suction as needed, and provide oxygen or assisted ventilation according to the patient's clinical condition.
- Check pulse and rhythm immediately when the presentation could represent cardiac arrest.
- Check glucose early, but do not delay midazolam during convulsive status epilepticus. Treat confirmed hypoglycemia under UP-4.
- Administer the age-appropriate midazolam regimen for seizure lasting 5 minutes or longer, unknown prolonged duration, recurrent seizure without recovery, or earlier when unstable airway, breathing, or circulation makes continued seizure immediately dangerous.
- For adult seizure persisting after the adequate midazolam regimen, administer ketamine 1 mg/kg IV/IO or IM, maximum 100 mg. Pediatric ketamine requires Medical Control authorization and dosing.
- Treat pregnancy or postpartum seizure under the obstetric/eclampsia protocol, trauma under the appropriate trauma protocol, and suspected toxin or withdrawal under the applicable toxicology protocol.
- Transport promptly with continuous reassessment and early receiving-facility notification for ongoing or recurrent seizure, incomplete recovery, or any high-risk feature.

Medications

Midazolam - Adult

10 mg IM preferred without IV/IO; 10 mg IN alternative; or 5 mg IV/IO when access is established

- For IV/IO administration, repeat 5 mg once after 5 minutes if seizure continues.
- Maximum total EMS midazolam dose: 10 mg.
- Paramedic only.

Midazolam - Pediatric

0.2 mg/kg IM/IN, max 10 mg; or 0.1 mg/kg IV/IO, max 5 mg

- May repeat once after 5 minutes if seizure continues.
- Maximum total: 0.4 mg/kg, not to exceed 10 mg.
- Count prescribed rescue benzodiazepine given before EMS arrival toward the two-dose maximum.
- Paramedic only.

Ketamine - Adult Refractory Seizure

1 mg/kg IV/IO or IM once; maximum 100 mg

- Use for seizure persisting after the adequate midazolam regimen.
- Continuous cardiac monitoring, SpO_2, waveform $EtCO_2$, and airway and ventilation readiness are required.
- Paramedic only.

Ketamine - Pediatric

Contact Medical Control for authorization and dosing

- Use only for seizure persisting after the adequate benzodiazepine regimen.
- Continuous cardiac monitoring, SpO_2, waveform $EtCO_2$, and airway and ventilation readiness are required.

Warnings / precautions

- Transport is strongly indicated for the listed high-risk features; those criteria alone do not establish incapacity or authority for involuntary transport. If a legally authorized decision-maker refuses, assess and document decision-specific capacity, explain material risks and alternatives, contact Medical Control for high-risk refusal, and follow the approved refusal/safeguarding procedure. Use an authorized emergency or protective-custody pathway when its legal conditions apply. A patient aged 16 or 17 remains a minor for consent purposes unless a legally recognized exception applies; the clinical age-16 dosing boundary does not establish legal adulthood.
- Do not forcibly restrain convulsions and do not place anything in the patient's mouth.
- Do not delay an indicated full anticonvulsant dose while attempting IV or IO access or obtaining glucose.
- Benzodiazepines and ketamine may produce apnea, hypoventilation, or airway obstruction; maintain immediate BVM, suction, and advanced-airway readiness.
- Brief seizure-like activity may occur at the onset of ventricular fibrillation or ventricular tachycardia; confirm pulse and rhythm.
- Postictal altered mental status does not exclude hypoglycemia, hypoxia, head injury, stroke, intoxication, infection, or eclampsia.

Clinical pearls

- Most isolated seizures stop within 1-2 minutes. A seizure lasting 5 minutes or recurrent seizures without recovery should be treated as status epilepticus.
- An adequate early benzodiazepine dose is more effective than several delayed, partial doses.
- Document the exact time seizure activity began, medication administration times, observed cessation, recurrence, and return to baseline.

- Postictal agitation should prompt reassessment for hypoxia, hypoglycemia, injury, hyperthermia, intoxication, and other medical causes before behavioral treatment.
- A focal deficit may be postictal, but persistent or unexplained focal findings require stroke or intracranial-emergency evaluation.

Special populations | Pregnancy and postpartum

- Treat seizure after 20 weeks of pregnancy or within 6 weeks postpartum as eclampsia until proven otherwise.
- Follow the obstetric/eclampsia protocol; magnesium sulfate is the primary anticonvulsant for suspected eclampsia.
- Do not allow benzodiazepine or ketamine administration to replace indicated magnesium therapy.

Special populations | Pediatrics

- Use pediatric protocols for patients younger than 16 years and calculate doses from actual or length-based weight.
- A first febrile seizure, prolonged or recurrent febrile seizure, incomplete recovery, abnormal examination, or age outside the expected simple-febrile-seizure range requires transport.
- Remove excess clothing and treat fever under the fever protocol after stabilization; do not use ice-water immersion or delay seizure treatment for cooling.

Special populations | Refusal after a seizure

- Transport is strongly indicated for first seizure, status or recurrent seizure, incomplete return to baseline, abnormal vital signs or glucose, injury, fever, pregnancy or postpartum state, persistent neurologic deficit, suspected overdose or withdrawal, respiratory compromise, or any EMS midazolam or ketamine administration.
- A capable adult with known epilepsy may refuse only after one typical seizure, complete return to neurologic baseline, normal examination, vital signs and glucose, no injury or concerning cause, no EMS anticonvulsant administration, and confirmation of responsible adult supervision.
- Complete the Refusal / Non-Transport protocol and document the patient's capacity, risks explained, responsible adult, and return precautions.

Source and references

Tennessee EMS BLS/ALS State Protocol Guidelines 2024-2025 - SOP 315 Seizures. | American Epilepsy Society - Evidence-Based Guideline: Treatment of Convulsive Status Epilepticus in Children and Adults, 2016. | RAMPART - Intramuscular versus Intravenous Therapy for Prehospital Status Epilepticus. | Scheppke et al. - Prehospital Ketamine for Benzodiazepine-Resistant Status Epilepticus, Critical Care Explorations, 2024. | Claiborne Covenant EMS Formulary - Midazolam and ketamine. | North Carolina College of Emergency Physicians UP-13 Seizure source protocol retained for historical comparison.

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